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Why does the Cochrane–Piazzesi model predict treasury returns?

Riccardo Rebonato [a,*], Ken Nyholm [b]

[a] EDHEC Risk Climate Institute - EDHEC Business School, 10 Fleet Place, London EC4M 7RB, UK
[b] European Central Bank - Risk Management, Sonnemannstrasse 20, 60314 Frankfurt am Main, Germany

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ABSTRACT

We explain why the Cochrane–Piazzesi (CP) model, which uses a single tent-shaped linear combination of forward rates, is so effective at predicting bond excess returns. By using a novel statistical test coupled with a popular resampling technique, first we rule out the possibility that the high predictability may be an artefact of in-sample overfitting. Then we find that, contrary to explanations proposed in the original CP paper, neither the specific tent shape of the factor loadings nor the four-to-five-year yield spread are essential for the model's predictive power. Instead, our analysis suggests that the predictive power of the CP model lies in its ability to identify the cointegration relationship among the quasi-unit-root forward rate regressors needed to produce the stationary process of excess returns. To support this interpretation we show that cointegration relationships among forward rates directly provide strong predictors of excess returns, and we propose that the cointegration modes of attraction generate at least part of the excess returns. Our findings shed new light on the source of bond return predictability captured by the CP factor and highlight the link between cointegration properties and the dynamics of yields.[1]

1. Introduction

The predictability of excess returns in US Treasuries has been a focal point of fixed-income research, with the first studies appearing as early as the late 1980s (Fama and Bliss, 1987; Campbell and Shiller, 1991). The field has witnessed a true renaissance in the new century with the work by Cochrane and Piazzesi (2005b), Cieslak and Povala (2015), Ludvigson and Ng (2009), Cooper and Priestly (2009) and many others. Among the new-generation models, the Cochrane and Piazzesi (2005b) approach is certainly the best known, and, arguably, the most successful in terms of predictive power. The Cochrane–Piazzesi (CP) factor refers to a single linear combination of forward rates, constructed by regressing average excess returns across maturities on a set of forward rates. The resulting regression coefficients typically display a distinctive 'tent shape' pattern when plotted against maturity: the regression coefficients are low at the shortest maturities, they peak at intermediate maturities, and are low again at the longest maturities. This creates a profile reminiscent of a tent, with the values of the coefficients forming a symmetric, peaked pattern.

Recent empirical studies (such as the work by Hellerstein (2011), Dahlquist and Hasseltoft (2013), Erikson (2017), or, more recently, Rebonato and Zanetti (2023)) have provided comprehensive empirical validation of the Cochrane–Piazzesi approach. Taken together, these studies confirm that the Cochrane–Piazzesi factor predicts better than the slope factor, both in sample and out of sample (since the Cochrane–Piazzesi factor has five non-intercept regression coefficients rather than a single one for the slope, the danger of overfitting must be considered seriously).

* Corresponding author.
E-mail addresses: riccardo.rebonato@edhec.edu (R. Rebonato), ken.nyholm@ecb.eu.europa (K. Nyholm).

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While the empirical fact that the Cochrane–Piazzesi model predicts well (and does so, as we have seen, also out of sample) is relatively well accepted, the explanation of *why* it does so is still imperfectly understood. In their original paper, Cochrane and Piazzesi (2005b) state that '[t]he four-to five-year yield spread, though a tiny factor for yields, provides important information about the expected returns of all bonds',[2] and add that '[t]o incorporate our evidence on risk premia, a yield curve model must include something like our tent-shaped return-forecasting factor in addition to such traditional factors as level, slope, and curvature, even though the return-forecasting factor does little to improve the model's fit for yields.'[3] They reinforce the point by stating that return-predicting factor 'is not fully captured by any of the conventional yield-curve factors. It reflects a four- to five-year yield spread that is ignored by factor models.'[4] So, in this interpretation, there is something essentially informative in the four-to-five-year spread, and it is this extra information that allows a better prediction than what is afforded by the slope. We show empirically, however, that this is not the case: when we extend the maximum maturity to ten years, a Cochrane–Piazzesi-like factor remains predictive of returns *even if the four-to-five-year spread is totally absent from the regressors*.

Along different lines, Campbell (2018) argues that '. . . the [tent] predictor measures the concavity of the yield curve between maturities one and five years. This has stimulated interest in equilibrium models whose yield curves are concave at times when risk premia are high. . . ',[5] and observes that '. . . the return predictor is cyclical, tending to increase during recessions when the Fed lowers the short-term interest rate and creates a yield curve that is steep between one and three years. This finding suggests the possibility that bond return predictability can be linked to cyclical predictability of other assets, as suggested by early empirical work by Fama and French (1993) and equilibrium asset pricing models with cyclical variations in risk or risk aversion.'[6] The emphasis of this explanation links the cyclicality of the Cochrane–Piazzesi factor to the cyclicality in risk aversion — an explanation that, as we shall show, has some similarity to the mechanism we propose. However, we shall also show that neither the steepness of the yield curve between one and three years nor the specific tent shape of the factor play any essential role.

What is then responsible for the predicting power of the Cochrane–Piazzesi return-predicting factor? First of all, since most empirical studies have been conducted in-sample, we want to confirm that it is not due to overfitting. To this end, we present a new test, more powerful than the one presented in Rebonato and Zanetti (2023). More precisely, we first regress real-word excess returns against real-world forward rates, and estimate the associated Cochrane–Piazzesi regression coefficients. We then resample thousands of yield curves using the procedure in Crump and Gospodinov (2019a); calculate excess returns and forward rates in the resampled world; *and then use regression coefficients estimated in the real world to predict returns in the resampled universes*. When we do so, we find only a small deterioration in explanatory power (as measured by the R^2 coefficient) compared to using coefficients determined by regressions carried out in the resampled worlds. This test combines the desirable feature of being fully out of sample with the much greater statistical power than what can be achieved by splitting one single real-world sample in a training and a testing set.

Having set to rest the possibility that the high predictability may be due to overfitting, we present our explanation of why the Cochrane–Piazzesi factor predicts in the first place. More precisely, first we show that, while the forward-rate regressors are quasi-unit-root processes, the left-hand variables, ie, the excess returns, are strongly mean-reverting, $I(0)$, processes. We then interpret the predictive regression that determines the loadings of the forward rates as a procedure to identify the cointegration relationship among the forward-rate regressors to turn their processes into an $I(0)$ process that can efficiently mimic the properties of the excess returns. In other words, the linear combination produced by the regression coefficients is (a rotation of) the cointegration relationship needed to produce a stationary process. If our conjecture is correct, the cointegration relationships among the forward rates should be good predictors of excess returns. We find that this is indeed the case.

This, however, is not the full story. It is, of course, not enough for a return-predictive factor to be stationary to predict excess returns. For predictability, the reversion speeds of the factor and of the excess returns must be similar, and the two processes must be 'in phase'. We explain in Section 5 that this happens because what we call the 'modes of attraction' that result from the cointegration relationships actually *generate* at least a part of the excess returns, and lend to these excess returns their own predictability. As we show, this allows us to draw interesting parallels among seemingly very different return predicting factors, such as the Cochrane–Piazzesi, the 'cycles' by Cieslak and Povala (2015), the slope and, of course, the cointegration factor that we create.

2. Literature review: Are the current explanations convincing?

In the introduction we have mentioned some of the explanations that have been proposed to account for the strong predictive power of the Cochrane–Piazzesi return predicting factor, and we have indicated that we find these explanations wanting. In the first subsection we present evidence from the literature that casts doubts on these prevailing explanations; in the following subsection we introduce original analyses aimed at identifying what can and what cannot account for the observed predictability.

[2] page 139
[3] *ibid.*
[4] page 146
[5] page 241
[6] *ibid.*

Table 1
The adjusted R^2 of the regression of the excess returns for initial maturities ranging from 2 to 10 years (left column) against the first five forward rates. Data frequency: Monthly. Sample period: June 1971–September 2023. Excess returns calculated for 1-year holding periods.

Initial maturity	R^2
2	0.120
3	0.125
4	0.137
5	0.150
6	0.164
7	0.176
8	0.186
9	0.194
10	0.201

2.1. Does the 'Tent' shape matter?

As we have seen, the 'beautiful tent shape' of the loadings on the five regressors in Eq. (5) is presented by Cochrane and Piazzesi (2005a) as being important to explain the predictive power of the return predicting factor. The first strand of criticism of this aspect of the Cochrane–Piazzesi model probably dates back to Dai et al. (2004), who point out that almost arbitrary choices for the interpolation method used to obtain the forward rate regressors can alter significantly the loadings on the regressors: They find that 'even small measurement errors, in the presence of highly correlated forward rates, can lead to large differences in fitted projection coefficients'.[7] The results by Villegas (2015) are particularly interesting in this respect: she starts from a set of data that do *not* produce a tent pattern (she uses the term 'bat pattern' for the shape of the loadings, a term that we shall use as well), and then adds a small amount of uninformative noise to the data. When the regression is run on the original signal plus noise, Villegas (2015) finds that the original bat often mutates into a tent. Further evidence that the precise tent pattern is not essential for high predictability is also provided in Rebonato (2018).

In sum: no special meaning should be ascribed to the specific tent shape of the factor, as different loadings, *as long of alternating signs*, give rise to factors with almost identical predictive power. The question, however, remains open as to why this alternating-sign pattern should have predictive power.

2.2. Does the 4-to-5 year curvature convey information?

As mentioned in the introduction, Cochrane and Piazzesi (2005a) argue that there is something special and particularly informative about excess returns in the curvature of the yield curve between the specific maturities of four and five years (but do not specify what this extra information is, or why it is conveyed exactly by the four-to-five-year curvature). Their original study only extended in maturities to five years and so, if they wanted to use five regressors, they could not avoid including the fourth and fifth forward rate. This restriction is however no longer present if one extends the maturity range. To check the validity of their explanation we have therefore extended the maximum initial maturity from five to ten years, and proceeded along two different lines. First, we have used the Cochrane–Piazzesi combination of the first five forward rates to predict returns for strategies with initial maturities as long as ten years. When we do so, we find that Cochrane–Piazzesi return-predicting factor predicts 10-year returns even better (with a higher adjusted R^2, 0.201 versus 0.150) than 5-year returns. See Table 1, which shows the adjusted R^2 of the regression of the excess returns for initial maturities ranging from 2 to 10 years (left column) against the first five forward rates. It is difficult to imagine how specific features of the four-to-five-year curvature can contain useful information about the 10-year returns.

The more decisive rebuttal of the conjecture that useful information is contained in the four- and five-year forward rates, however, comes from the flexibility in choosing the five regressors afforded by extending the longest maturity (and hence the maturity of the forward rates) to ten years. By doing do, we can now choose combinations of forward rates that do not include the four-

[7] page 7. Cochrane and Piazzesi (2004) provide a rebuttal, which is centred on the claim that the smoothness of the Nelson and Siegel (1987) procedure (which underlies the approach by Gurkaynak et al. (2007)) irons away not just noise, but also useful information: in their words, it "throws away the baby with the bathwater".

and five-year maturities at all. When we do so, we find that the predictive power (again, measured as the R^2 of the regression) varies relatively little with the choice of five forward rates out of ten, and that including the fourth and fifth forward rate does not produce significantly higher explanatory power. More precisely, there are $\binom{5}{10} = 252$ ways to choose five forward rates out of ten, and we have calculated the R^2 coefficients[8] for every single one of them. We have then sorted the combinations of forward rates in order of ascending R^2. We found that none of the top 37 R^2 coefficients were generated by combinations of forward rates that included the 4-year and 5-year-maturity forward rates (ie, approximately, combinations of forward rates which include the 4-year and 5-year-maturity forward rates do not generate any of the top 15% R^2 coefficients). Conversely, the 4-year and 5-year-maturity combination produces 3 of the 10 *lowest* R^2 coefficients.

On the basis of these observations, we can conclude that no special information appears to be contained in the four-to-five-year curvature. In the rest of the paper we therefore focus on finding an alternative explanation.

2.3. Other predicability studies using cointegration

Another strand of the literature argues for the existence of cointegration in yields from a macro-based viewpoint. Bauer and Rudebush (2020), for example, highlight the stochastic nature of macroeconomic trends and their impact on interest rates, arguing for the importance of allowing for shifting long-run trends. They show that stochastic trends in macroeconomic variables are important for understanding the persistence of interest rates and their predictive power for bond returns and find that these trends provide predictive gains over traditional models. Favero et al. (2024) employ a dynamic term structure model incorporating time-varying macroeconomic trends like trend inflation and equilibrium real interest rates to explain the term structure of interest rates. They demonstrate that incorporating these trends significantly improves the understanding and prediction of interest rates and bond risk premia. In contrast, our approach emphasizes the micro-level dynamics of forward rates. By linking the CP model's effectiveness to cointegration, we offer a novel interpretation that complements the macro-based insights of, among others, Bauer and Rudebush (2020), Favero et al. (2024).Berardi et al. (2021) provide an insightful perspective on bond return predictability by introducing the convergence gap defined as the difference between the natural rate of interest and the real federal funds rate, capturing the impact of real economic imbalances on interest rates. Specifically, the convergence gap 'cleans up systematic patterns in forecast errors related to the state of the economy. This is achieved by eliminating the predictability of residuals, thus enhancing the predictability of bond returns when included alongside forward rates in regression models.' In contrast we focus on the role of cointegration among forward rates in predicting bond returns. This cointegration transforms non-stationary forward rates into a stationary process, effectively 'cleaning up' the bond return predictor by filtering out noise related to future interest rate movements. Both approaches underscore the importance of integrating economic and statistical insights to improve the forecasting of bond returns. While we highlight the statistical properties of forward rates, Berardi et al. (2021) emphasize the economic implications of monetary policy deviations captured by the convergence gap. In fact, the convergence gap can be seen as an observable variable that complements the empirical cointegration factor by addressing economic imbalances and monetary policy deviations, thus enhancing predictability. Together, these approaches highlight the importance of integrating economic and statistical insights to enhance bond return forecasting.

3. Descriptive statistics

Yield curve data from the US market form the basis for our empirical investigation. Zero-coupon yields from Gurkaynak et al. (2006) covering the period from June 1961 to September 2023 are sampled at a monthly frequency and used to calculate forward rates and excess returns. As mentioned, these have been obtained using the well-known (Nelson and Siegel, 1987) fitting model. Since their approach and its variants have been widely discussed in the literature, we do not dwell on this point, and direct the reader to Pooter (2007) for a recent review.[9]

We denote the price observed at time t of an n-year discount bond as $P_t^{(n)} = \exp(-n \cdot y_t^{(n)})$, with the yield defined as:

$$y_t^{(n)} \equiv -\frac{1}{n} p_t^{(n)}, \tag{1}$$

where $p_t^{(n)}$ is the log price of the n-maturity discount bond. The time-t forward rate between time $t+n-1$ and $t+n$ is then:

$$f_t^{(n)} \equiv p_t^{(n-1)} - p_t^{(n)}, \tag{2}$$

and the holding period return from buying an n-year bond at time t and selling it at time $t+1$ is:

$$r_{t+1}^{(n)} \equiv p_{t+1}^{(n-1)} - p_t^{(n)}, \tag{3}$$

with the excess return defined as:

$$rx_{t+1}^{(n)} \equiv r_{t+1}^{(n)} - y_t^{(1)}. \tag{4}$$

[8] Unless otherwise stated, in what follows mention of the R^2 coefficients should always be understood to refer to *adjusted* coefficients.

[9] It is highly unlikely that our empirical results are dependent on the specific dataset used, and it is improbable that our conclusions would change materially if a different dataset, such as the Fama–Bliss data or the Liu–Wu data (Liu and Wu, 2021), were employed. Our findings rely on the presence of cointegration among rates within the sample, a characteristic consistently demonstrated in yield curve data samples in many currencies. This feature is also clearly evident in data provided by commercial sources like Bloomberg and Datastream.

Table 2
Description of the data.

	Half life	mean	stdev	adf test	kpss test
$f^{(1)}$	6 y 6 m	4.83	3.32	I(1)	I(1)
$f^{(2)}$	7 y 4 m	5.24	3.24	I(1)	I(1)
$f^{(3)}$	8 y 11 m	5.54	3.11	I(1)	I(1)
$f^{(4)}$	9 y 3 m	5.77	2.98	I(1)	I(1)
$f^{(5)}$	8 y 11 m	5.96	2.88	I(1)	I(1)
$f^{(6)}$	8 y 6 m	6.12	2.80	I(1)	I(1)
$f^{(7)}$	7 y 11 m	6.24	2.75	I(1)	I(1)
$f^{(8)}$	7 y 4 m	6.34	2.71	I(1)	I(1)
$f^{(9)}$	6 y 10 m	6.41	2.68	I(1)	I(1)
$f^{(10)}$	6 y 3 m	6.47	2.67	I(1)	I(1)
$xret^{(2)}$	11 m	0.40	1.67	I(0)	I(0)
$xret^{(3)}$	10 m	0.70	3.04	I(0)	I(0)
$xret^{(4)}$	11 m	0.94	4.22	I(0)	I(0)
$xret^{(5)}$	10 m	1.15	5.30	I(0)	I(0)
$xret^{(6)}$	10 m	1.31	6.31	I(0)	I(0)
$xret^{(7)}$	10 m	1.44	7.28	I(0)	I(0)
$xret^{(8)}$	9 m	1.55	8.23	I(0)	I(0)
$xret^{(9)}$	9 m	1.63	9.16	I(0)	I(0)
$xret^{(10)}$	9 m	1.68	10.08	I(0)	I(0)

The table reports the statistical properties of the data used in the empirical analysis of the paper. $f^{(n)}$ denotes the forward rate for maturity n and $xret^{(n)}$ denotes the excess return above the 1-year yield. The column labelled 'Half life' displays in years and months the speed at which the variables revert to their mean or equilibrium value. These values are calculated assuming the series follow an $AR(1)$ process: $\mathbb{E}[x_{t+h}] = \phi^h \cdot x_t$, so for the half life we have $\frac{1}{2}x_t = \phi^h \cdot x_t$, and $h = \frac{log(\frac{1}{2})}{log(\phi)}$. The columns labelled 'mean' and 'stdev' show the two first moments of the distributions of the variables. To test for stationarity, we rely on Dickey and Fuller (1979) and Kwiatkowski et al. (1992). We report the results of the tests in the last two columns of the table. The table contains the results of the tests at a 5% confidence level. $I(0)$ indicates that the variable is stationary, i.e., integrated of order 0 and $I(1)$ indicates that the series is integrated of order 1.

Table 3
Correlation structure of the data.

	Forward rates	Excess returns
PCA1	96.2	98.5
PCA2	3.4	1.4
PCA3	0.4	0.1
PCA4	0.0	0.0

The table shows the variance explained by the first four principal components (PCAs) extracted from the forward rates and the excess returns.

Our sample of forward rates comprises 748 monthly observations for $n = \{1, 2, \ldots, 10\}$ years, while the sample of annual returns in excess of the 1-year yield comprises 736 monthly observations for $n = \{2, 3, \ldots, 10\}$. Table 2 summarizes the statistical properties of these data, and Table 3 illustrates the strong cross correlation among the data series.

As discussed in Section 2.2, the original Cochrane–Piazzesi analysis was carried out for yield maturities out to five years, but in our study we extend the maximum maturity to 10 years. In the Cochrane–Piazzesi study the associated return predicting factor, $RPF^{(n)}_{CP} = \alpha^{(n)} + (\beta^{(n)})^\top \boldsymbol{f}_t$, is calculated by regressing each of the four n-maturity excess returns against the five forward rates:

$$rx^{(n)}_{t+1} = \alpha^{(n)} + \left(\beta^{(n)}\right)^\top \boldsymbol{f}_t + \epsilon^{(n)}_{t+1}. \tag{5}$$

Following Cochrane & Piazzesi (2005), we use the restricted CP factor obtained from regressing on forward rates the average excess returns, defined as

$$\frac{1}{4}\sum_{n=2}^{5} rx^{(n)}_{t+1} = \overline{rx}_{t+1}.$$

The maturity-specific factors (unrestricted model) are used only for comparative analysis in the context of slope regressions.

When the yield data used by Cochrane and Piazzesi (2005a) are used, for each maturity the loadings $\beta^{(n)}$ then arrange themselves in the shape of a 'tent', as shown in left-hand panel of Fig. 1. As discussed in Section 2.1, with other yield data different shapes are recovered, of which the 'bat' is the more common (see the left-hand panel of Fig. 1).

In our explanation of the reason for the high predictive power of the Cochrane–Piazzesi the non-stationary behaviour of the regressors and the $I(0)$ character of the left-hand variable (the excess returns) play an essential role. We therefore look at these aspects in some detail.

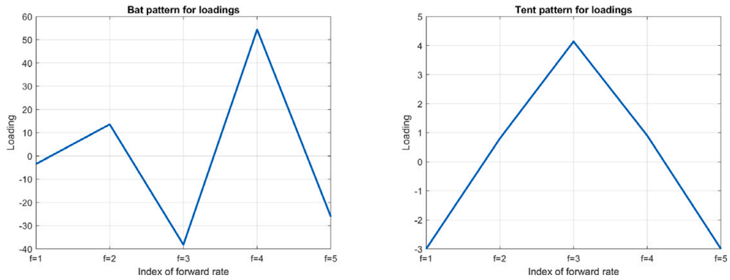

Fig. 1. The 'bat-shaped' loadings $\beta^{(n)}$ for $n = 5$ obtained with the yield data from Gürkaynak and Wright (2012) (left panel), and the same quantities arranged in the shape of a 'tent', obtained with the yield data used by Cochrane and Piazzesi (2005a) (right panel).

Clear evidence for the non-stationary behaviour of forward rates is presented in Table 2. Forward rates of all ten maturities are identified as integrated of order one, denoted as $I(1)$, at a significance level of 5% using the Augmented Dickey–Fuller test (Dickey and Fuller, 1979) and the KPSS test (Kwiatkowski et al., 1992).[10] The calculated half-lives, also displayed in Table 2, further illustrate the statistical properties of the forward rates. Assuming that they follow an autoregressive process of order one, $f_t^{(n)} = c + \phi \cdot f_{t-1}^{(n)}$ for all n, we cannot statistically rule out that $\phi = 1$, but we find the sample estimate of ϕ to be less than 1 for all series, as evidenced by the fact that the calculated half-lives are all finite. Still, all forward rates are extremely slowly mean reverting, with half-lives extending to decades.

This non-stationary behaviour should be contrasted with that of excess returns, which exhibit stationarity for all maturities, as also shown in Table 2. The performed statistical ADF and KPSS tests confirm this at the 5% level of significance. Calculated half-lives are all below one year, corroborating the conclusions from the statistical tests. For future reference, we note that the speed of reversion towards the mean of the excess return series is similar to the one-year holding period that we use in the empirical studies reported in the following.

Table 3 shows that a few principal components explain the majority of the variance matrices of the forward rates and the excess returns. This is a well documented phenomenon that has permeated term structure models since the three-factor structure of Nelson and Siegel (1987) and the work by Litterman and Scheinkman (1991).

For future reference, we also calculated the regression of the excess returns against the yield curve slope, defined as the difference between the 10-year and the 1-year yields:

$$rx_{t+1}^{(n)} = \alpha_{sl}^{(n)} + \beta_{sl}^{(n)} \cdot slope_t + \epsilon_{sl,t+1}^{(n)}. \tag{6}$$

Fig. 2 displays the Cochrane–Piazzesi and the slope return-predicting factors (left-hand panel) and the difference between the two (right-hand panel). It is important to note how highly correlated the two factors are, and that the difference from the two factors (shown in the right-hand panel of Fig. 2) displays a strongly mean-reverting behaviour. So, the Cochrane–Piazzesi factor recovers the information captured by the slope, and adds a mean-reverting signal to it. The degree of correlation between the slope and the Cochrane–Piazzesi factors is remarkable if one considers how different the regressors are. We explain in Section 5 that this high correlation, far from being a coincidence, stems from the fact that the slope factor as we have defined it can be seen as a particularly simple cointegration relationship between the first and tenth forward rates.

Finally, we point out, also for future reference, that, when we regress the residuals, $\epsilon_{sl,t+1}^{(n)}$, of the slope regression in (6) against the Cochrane–Piazzesi factor, we find highly significant intercept and slope. When the opposite test is done, ie, when we regress the residuals of the Cochrane–Piazzesi regression against the slope, the resulting regression coefficients are statistically indistinguishable from zero. In other terms: the Cochrane–Piazzesi factor completely 'drives out' the slope in a predictive regression of excess returns.

4. Cointegration relationships as return predictors

As shown in Section 3, forward rates are integrated of order 1 and excess returns are stationary. As explained in the introductory section, our contention is that the regression in Eq. (5) finds the coefficients necessary to turn the $I(1)$ processes of the forward rate regressors into an $I(0)$ process that closely tracks the $I(0)$ process of the excess returns. A necessary condition for our conjecture to be true is that it should be possible to find a cointegration relationships between the forward rates.[11] We therefore estimate cointegration relationships among the forward rate using the standard set up:

$$\Delta f_t = c + \alpha \cdot \left(\beta^\top \cdot f_{t-1}\right) + \sum_{j=1}^{J} B_j \cdot \Delta f_{t-j} + e_t. \tag{7}$$

[10] We note in passing that in a well-functioning economy, forward rates cannot truly be $I(1)$ processes, as they must be constrained within finite bounds, or economic activity would cease, and the economy would collapse. However, within the sample and from a purely econometric perspective, it is reasonable to model forward rates as $I(1)$ processes.

[11] In this context it is important to observe that $\Pi = \alpha \cdot \beta^\top$ is not uniquely identified and that any proper rotation matrix can alter their interpretation, since $\alpha \cdot \beta^\top = \alpha \cdot I \cdot \beta^\top = \alpha \cdot A^{-1} \cdot A \cdot \beta^\top$, where I is the identity matrix and A a rotation (and hence invertible) matrix.

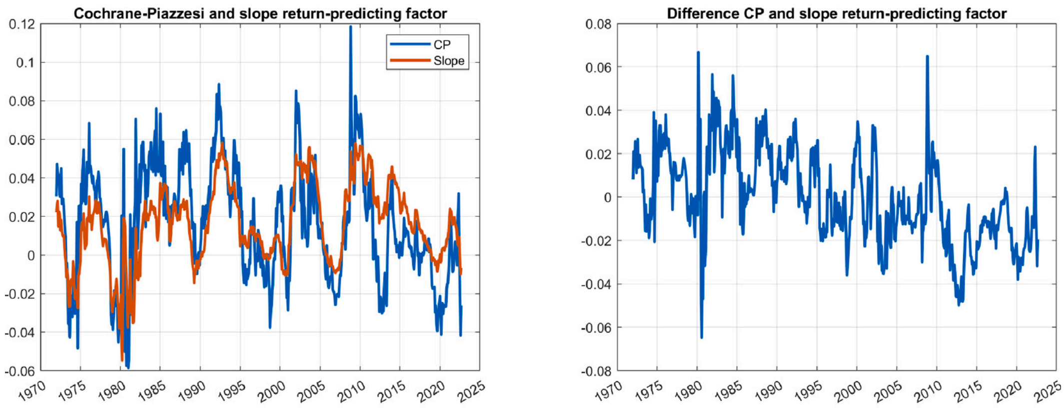

Fig. 2. The Cochrane–Piazzesi (blue line) and the slope (red line) return-predicting factors (left panel) and the difference between the two (right panel). The Cochrane–Piazzesi factor is given by the fitted value of a regression of average excess returns on all forward rates.

Table 4
Cointegration analysis for the first 5 forward rates.

r	h	stat	cValue	pValue
0	1	286.24	76.97	0.0010
1	1	139.50	54.08	0.0010
2	1	62.27	35.19	0.0010
3	1	21.40	20.26	0.0347
4	0	3.50	9.16	0.5582

The table shows the results of the Johansen cointegration test applied to the data set of forward rates. The trace-test is performed at a 5% level of significance for the null hypothesis of cointegration rank less than or equal to r. h indicates whether the null is accepted $h = 1$, or is rejected $h = 0$. The tests show that a cointegration rank of $r = 4$ is identified with a $pValue = 55.82\%$.

If the matrix $\Pi = \alpha \cdot \beta^{\top}$ has less than full rank, then $x_t = \beta^{\top} \cdot f_{t-1}$ represents a linear, and stationary, combination of the non-stationary forward rates, ie, the long-run cointegration relationship among the variables, and α can be interpreted as the speed at which perturbed data converge to this equilibrium.

We estimate Eq. (7) using the Johansen method, as described in Johansen (1988, 1991). To mimic as closely as possible the study in Cochrane and Piazzesi (2005b), we first use forward rates observed at maturities $n = \{1, 2, 3, 4, 5\}$. Table 4 shows that a cointegration rank of $r = 4$ is identified for the forward rates. This means that there are four cointegrating relationships among the first five forward rates. Since in our study we extend analysis to a maximum maturity of 10 years, we also carry out a cointegration analysis for all ten forward rates. Table 5 shows that at the 5% significance level $r = 9$ cointegration relations are detected.

One may ask how robust the cointegration relationships are over different periods. To answer this question, we have carried out the following test: (i) we have selected forward rates from the period from 1961 to Dec 1982; (ii) we have estimated the cointegration factors as explained above; (iii) we have stored these cointegration factors; (iii) we have expanded the data with an additional 12 months of observations; and (v) we have repeated the steps (i) to (iv) until the end of the data. The results for the first and the second cointegration factors are shown in Fig. 3. While the lines share some common data, by the end of the test, almost twice as much data as what was originally used has been added to the initial set (we use overlapping data to facilitate visual comparison.) It is clear that the cointegration relationships we estimate are extremely stable.

In sum: we have so far shown that the regression of the five or ten forward rates describing a 'short' or 'long' yield curve are $I(1)$ processes; that all the excess returns are $I(0)$ processes; and there exist four and nine cointegration relationships among the five and ten regressors, respectively. We investigate more systematically in what follows the conjecture that cointegration and the explanatory power of the Cochrane–Piazzesi factor may be linked.

5. The empirical analysis and results

With our empirical analysis we test two separate aspects of the Cochrane–Piazzesi model: first, in view of the caveats presented by Bauer and Hamiltom (2018), and since most analyses so far have been in-sample, we explore whether its high predictability is

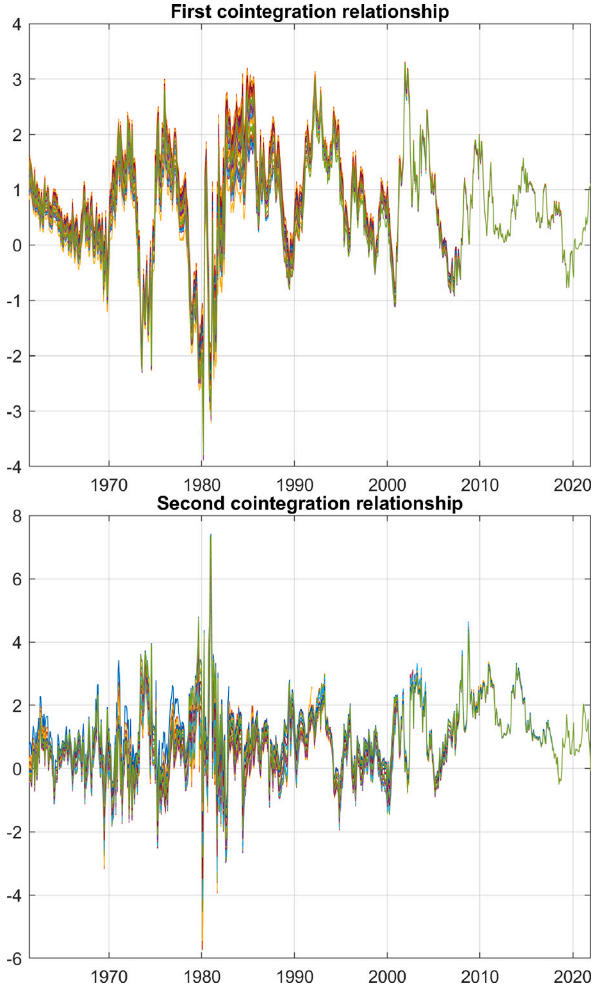

Fig. 3. The first and second cointegration factors (top and bottom panel, respectively) calculated using steps (i) to (v) described in the text.

Table 5
Cointegration analysis for all forward rates.

r	h	stat	cValue	pValue
0	1	1284.70	251.26	0.0010
1	1	942.23	208.44	0.0010
2	1	675.38	169.60	0.0010
3	1	482.79	134.68	0.0010
4	1	322.83	103.85	0.0010
5	1	202.95	76.97	0.0010
6	1	120.83	54.08	0.0010
7	1	59.55	35.19	0.0010
8	1	20.78	20.26	0.0426
9	0	2.92	9.16	0.6448

The table shows the results of the Johansen cointegration test applied to the data set of forward rates. The trace-test is performed at a 5% level of significance for the null hypothesis of cointegration rank less than or equal to r. h indicates whether the null is accepted $h = 1$, or is rejected $h = 0$. The tests show that a cointegration rank of $r = 9$ is identified with a p value of 64.48%.

due to over-fitting, or to its ability to capture information about excess returns which is missed by other return-predicting factors. Once we have established that overfitting is not responsible for the high predicting power of the factor, we want to see whether this extra information indeed comes from creating a particular cointegration relationship among forward rates capable of reproducing the (stable) statistical characteristics of the excess-return processes. We start with an investigation of whether the high predictability is due to overfitting.

5.1. Out-of-sample testing for overfitting

To ruled out overfitting, we conduct a very demanding non-parametric resampling test, based on the idea of estimating regression coefficients in the real world, using them in the resampled universes, and studying the deterioration of predictability that results from this ‘scrambling of coefficients’. More precisely, we proceed as follows.

- At the beginning of every month in the period described in Section 3 we buy a T-maturity discount bond, financing its purchase by the sale of a 1-year discount bond, with T equal to $T = 2, 3, \ldots, 10$. We hold the position for one year, at which time we sell the now $T - 1$-year maturity bond.
- We calculate the excess returns in the real world using (4) for the nine initial maturities. We also calculate the average excess return (averaged over initial maturities).
- We regress the nine excess returns thus calculated and the average excess returns against five forward rate regressors (in-sample analysis), and estimate the coefficients of the real-world Cochrane–Piazzesi return-predicting factor.
- We resample the yield curve history using the non-parametric approach described in Crump and Gospodinov (2019a). (We carry out 10,000 resamplings). The sample size of the resampled data is kept identical to the original data set consisting of $n = \{1, 2, \ldots, 10\}$ maturities and 748 monthly observations.[12]
- For each resampled universe, we calculate the loadings of the Cochrane–Piazzesi return-predicting factor associated with that particular resampled universe.
- We estimate two sets of R^2 coefficients of determination:

  1. the first, denoted by the symbol R^2_{ss}, is obtained by regressing the excess returns obtained in the simulation against the Cochrane–Piazzesi factor obtained in the same simulation;
  2. the second, denoted by the symbol R^2_{rs}, is obtained by regressing the excess returns obtained in the simulation against an ‘out-of-sample’ Cochrane–Piazzesi factor obtained by combining the forward rates realized in the simulation and the same loading coefficients obtained *in the real world*.

- We compare the distribution of R^2 coefficients thus obtained.

The results of the comparison are shown in Fig. 4, which displays the surprisingly similar distributions of the R^2_{ss} and R^2_{rs} coefficients. The difference between the means of the two distributions is statistically significant, but we note that, for a finite sample, the excess return predictability estimated using the resampled data is always biased downward. This happens because, when two segments of the yield curve in the Crump and Gospodinov (2019a) resampling are joined together, no predictability for the first twelve excess returns in the ‘new’ segment can be expected from the regressors belonging to the last twelve month of the ‘old’ segment. In Appendix A we measure this bias by estimating via Crump and Gospodinov (2019a) resampling the R^2 for synthetic data for which we know what the degree of predictability should be. When we do this for sample sizes and lengths of the resampling segments similar to what we have in our study, we find the bias to be approximately 15-to-20%. When we correct the mean of R^2_{rs} by this amount, we find that it becomes statistically indistinguishable from the mean of the R^2_{ss} coefficients.

This test mimics the out-of-sample strategy of estimating the Cochrane–Piazzesi using forty years of data, and then using (10,000 times) these coefficients without change for the next forty years. The near-coincidence (after bias corrections) of the means of R^2_{rs} and R^2_{ss} provides a strong indication that the good predictive ability of the Cochrane–Piazzesi factor does not come from over-fitting.

Interesting and encouraging as these results are, they still give no indications about the reason for this impressive performance. We therefore move to our analysis of the cointegration conjecture.

5.2. Testing the cointegration conjecture

The first part of our explanation of why the Cochrane–Piazzesi factor is so effective is that it finds the coefficients necessary to turn the non-stationary forward-rate regressors into an $I(0)$ process with similar mean-reversion properties as the excess returns. In effect, in our explanation the loadings of the Cochrane–Piazzesi factor are cointegration coefficients (as explained, not uniquely defined, because only identifiable to within a rotation.) If this is correct, the cointegration factors themselves should provide reasonable regressors for the excess returns. We test this hypothesis as follows.

First, we note that augmented Dickey–Fuller tests confirm that the Cochrane–Piazzesi and cointegration return prediction factors behave as stationary processes. For both, the null hypothesis of the presence of a unit root is rejected with p-values of 0.0032 for the Cochrane–Piazzesi factor and of 0.0027 for the cointegration factor. Furthermore, we find that the half-life of the Cochrane–Piazzesi

[12] A block-sample size of 68 observations is used requiring 11 samples to obtain 748 observations. The block length was chosen using the correction reported in Patton et al. (2009) to the original procedure presented in Politis and White (2004). Data are resampled from the original data wrapped around a circle, as proposed by Crump and Gospodinov (2019b).

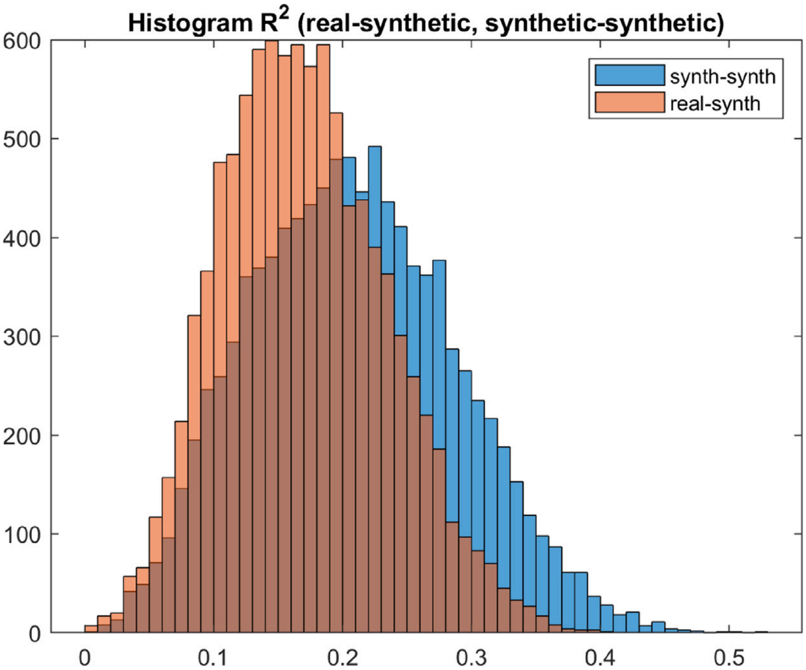

Fig. 4. The histogram of the R^2 coefficients obtained consistently using the resampled universes (blue bars, histogram to the right, labelled 'synth-synth'), and the histogram of the R^2 coefficients estimated in the real world and applied to the forward rates obtained in the resampled universes (brown bars, histogram to the left, labelled 'real-synth').

and the cointegration factors are 12 and 10 months, respectively. These estimates compare well to the half-life of the excess returns reported in Table 2.

To test our conjecture further, we systematically build in steps more and more complex cointegration return-predicting factors. More precisely, we compare and contrast how we build the cointegration factors with how one builds a generalized Cochrane–Piazzesi factor – a procedure with which the reader is certainly familiar (by a *generalized* Cochrane–Piazzesi procedure we mean a regression which has excess returns as left-hand variables, and n forward rates as right-hand regressors, with n not necessarily equal to 5 – when $n = 5$ one recovers, of course, the exact Cochrane–Piazzesi method). The reason for describing the two procedures starting from the case of two forward rates is that the cointegration relationship is in this case particularly simple, and maps almost exactly on the well-known slope factor, of which it therefore offers a different interpretation.

We start from two forward rates ($n = 2$). When these are used as regressors for excess returns in a generalized Cochrane–Piazzesi procedure, two 'slope' regression coefficients are estimated, and we find that these two slope coefficients put similar-magnitude, opposite-sign, weights on the two forward rates. Next, still with two forward rates, we move to the cointegration-based regression. More precisely, we estimate first the cointegration relationship among the two forward rates. Then we use the resulting linear combination of rates as the single return-predicting factor in a regression with excess returns as left-hand variables. From this regression we estimate a single 'slope' coefficient. Empirically, we find that the loadings on the two forward rates obtained using the generalized Cochrane–Piazzesi procedure and using the cointegration regression are in essence the same: for instance, choosing forward rates 2 and 4, we obtain very similar ratios of -1.15 and of -1.06 for the two-forward-rate regression and for the cointegration coefficients, respectively. We note in passing that the traditional Fama–Bliss-like slope factor is effectively doing the same thing, by assigning a $+1$ and -1 weight to the front and back and the yield curve. This explains the high degree of correlation between the forward-rate and the slope return-predicting factors, shown in the left panel of Fig. 2: the weights of $+1$ and -1 that create the slope are just approximate cointegration coefficients that combine the $I(1)$ processes for the first and tenth forward rate into a stationary process. Not surprisingly, the R^2 coefficients are all similar: 0.152 for the two-factor Cochrane–Piazzesi model, 0.137 for the cointegration model, and 0.102 for the slope.

Now we add one forward rate, ie, we consider the case when $n = 3$. With the generalized Cochrane–Piazzesi approach there is an extra forward rate regressor in addition to the two that we had before, and the excess-return regression yields three 'slope' coefficients. Clearly, if one compared the unadjusted R^2 coefficient from this regression with the R^2 coefficient obtained in the $n = 2$ case, this could not go down. As before, we now move to the cointegration regression. First, we establish a cointegration relationship among the three forward rates, and note that the cointegration coefficients are totally different from the $n = 2$ case, with two weights positive and one negative. These weights describe how *three* forward rates 'stay together' — in this sense, they describe a different 'mode of attraction', distinct from the mode of attraction between two forward rates, which consisted of two opposite-sign, similar-magnitude coefficients. ('By 'mode of attraction' we mean any mechanism, such as, for instance, a reversion speed matrix, that pulls

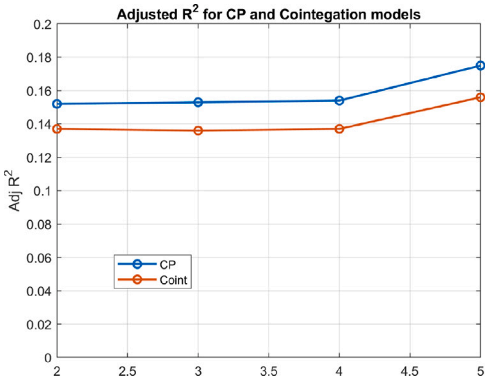

Fig. 5. The adjusted R^2 coefficients obtained using either the Cochrane–Piazzesi regression (line labelled 'CP') or the cointegration relationship (line labelled 'Coint') as return-predicting factor for the number of forward rates displayed on the x axis.

Table 6
The adjusted R^2 coefficients obtained using either the Cochrane–Piazzesi regression or the cointegration relationship as return-predicting factor for the number of forward rates in the leftmost column.

No of forward rates	R^2 (Cochrane–Piazzesi)	R^2 (Cointegration)
2	0.152	0.137
3	0.153	0.136
4	0.154	0.137
5	0.175	0.156

forward rates together when they drift apart.) If we use only the cointegration relationship obtained using three forward rates to predict excess returns, the R^2 coefficients plummets to 0.05. This happens because we are now using a different 'mode of attraction' among forward rates, and we have lost the mode of attraction that had been established with two forward rates. So, what we do instead is to consider $n - 1 = 2$ return predicting factors: the first is equal to the mode of attraction associated with two forward rates, and the second with the mode of attraction associated with three forward rates. We therefore determine by regression against excess returns two slope coefficient, each one associated with a different mode of attraction (cointegration relationship). The key point is that, to have a fair comparison, one has to *add* to the return-predicting factor arising from cointegration with two forward rates the new cointegration relationship obtained with three forward rates. As it happens, the new one adds very little, but so does adding a third forward rate to the original two in the generalized Cochrane–Piazzesi approach. Note that with the cointegration approach we still use only three forward rates (as many as in the Cochrane–Piazzesi approach).

This gives us a systematic procedure to add forward rates for the two sets of regressions. When we get to five forward rates, the R^2 for the generalized Cochrane–Piazzesi return-predicting factor climbs up to 0.175, and the R^2 obtained by using five 'modes of attraction' has a similar value of 0.156. Again, we stress that we are using the same number of forward rates for the two approaches. Table 6 shows the adjusted R^2 coefficients obtained using either the Cochrane–Piazzesi regression or the cointegration relationship as return-predicting factor for the number of forward rates in the leftmost column. Fig. 5 then shows how strongly correlated the two sets of R^2 coefficients are, and, in particular, the marked jump in R^2 as a fifth forward rate is added either to the Cochrane–Piazzesi or to the cointegration model.

This comparison between our cointegration procedure and the generalized Cochrane–Piazzesi method brings about an interesting observation. In their paper, Cochrane and Piazzesi (2005a) introduce a 'restricted' model, in which a tent shape is first posited for the five forward-rate regressors, and a single maturity-dependent slope coefficient is then 'fine-tuned' to obtain the least-square best return-predicting factor. With our cointegration procedure, we do something similar, because we use n forward rates, linked by $n - 1$ cointegration relationships, to create return predicting factors, and then obtain the slope coefficients that best account for the observed excess returns. There is an important difference, however: the tent-shaped return-factor in the Cochrane–Piazzesi procedure is arrived at by previous training on the existing excess return data, and its justification is purely empirical. In our approach, the cointegration relationships are conjectured on financial grounds to be good predictors, *and are estimated without using any information on excess returns* (information that goes instead into the construction of the 'tent' in the restricted Cochrane–Piazzesi method).[13]

[13] The results presented in Table 6 are for one particular choice of combination of forward rates. As in our analysis of the Cochrane–Piazzesi model discussed in Section 2.2, we find that the results depend modestly on the exact choice of forward rates. This again points to the fact that what matters for prediction is not including a particular-maturity forward rate, or a combination of particular-maturity forward rates, but capturing the relationships (what we have called the 'modes of attraction') among a given *number* of forward rates. Since the forward rates are highly correlated, for a fixed n, the behaviour of any n-tuple of forward rates is very similar.

The results presented in Table 6 have been obtained using real-world data. This leaves open the question of the statistical significance of the results. In order to gain statistical information about the predictive power of two regression models, we have employed a version of the Crump and Gospodinov (2019a) resampling procedure that can be described as follows.

1. As described in Section 5.1, we resample the (Gurkaynak et al., 2006) data using the approach suggested by Crump and Gospodinov (2019b). Again, the sample size of the resampled data is kept identical to the original data set consisting of $n = \{1, 2, \dots, 10\}$ maturities and 748 monthly observations. A block-sample size of 68 observations is used requiring 11 samples to obtain 748 observations. Data are resampled from the original data wrapped around a circle, as proposed by Crump and Gospodinov (2019b).
2. We calculate excess returns and forward rates using the resampled data.
3. We implement the approach of Cochrane and Piazzesi (2005b):

   3a. We run the regression[14]: $\frac{1}{4}\sum_{n=2}^{5} rx_{t+1}^{(n)} = \bar{rx}_{t+1}^{(n)} = \gamma^{\top} \cdot f_t + \bar{\epsilon}_{t+1}$, to identify the Cochrane–Piazzesi return predicting factor $\{\gamma^{\top} f\}_t$, ie, the linear combination of the forward rates that best predict the excess returns one year ahead. The vector f comprises forward rates of maturities $n = \{1, 2, 3, 4, 5\}$ years.

   3b. We run the regressions: $rx_{t+1}^{(n)} = b_n \cdot \{\gamma^{\top} f\}_t + \epsilon_{t+1}^{(n)}$, and record the obtained R^2 statistics from each regression for $n = \{2, 3, \dots, 10\}$.

4. We now use the cointegration relationships as return-predicting factors:

   4a. We apply the rank test to find the number of cointegration relationships among the forward rates using Eq. (7). Cointegration relations are found for a progressively increasing number of forward rates, starting with two, then three, four, and finally five. We denote them by $CI_t^k = \{\beta^{\top} f\}_t^{(k)}$, where $k = \{1, 2, \dots, r\}$ indicates the kth cointegration relationship.

   4b. We run a regression similar to step $3a$ using the identified cointegration relationships in step $4a$ instead of the forward rates directly: $\frac{1}{4}\sum_{n=2}^{5} rx_{t+1}^{(n)} = \bar{rx}_{t+1}^{(n)} = \gamma_{CI}^{\top} \cdot CI_t + \bar{\epsilon}_{t+1}$, to identify the cointegration-based return predicting factor $\{\gamma_{CI}^{\top} \cdot CI_t\}_t$, ie, the linear combination of the cointegration factors that best predict the excess returns one year ahead.

   4c. We run the regressions: $rx_{t+1}^{(n)} = v_n \cdot \{\gamma_{CI}^{\top} \cdot CI_t\}_t + \eta_{t+1}^{(n)}$. Since we are using the five first forward rates to identify the relevant cointegration relationships, there can be a maximum of 10 right-hand-side variables. This happens if all of the tested forward rate combinations have a ranks of $n-1$. To avoid potentially over-fitting, we rely on a stepwise linear regression approach where the best fit is found among the fewest right-hand-side variables. Even though the return prediction factor is trained on the average return between the first five excess returns, we perform regressions for excess returns of maturity $n = \{2, 3, \dots, 10\}$. $\{\gamma_{CI}^{\top} \cdot CI\}_t$ is the vector of observations for the linear combination of the cointegration relationships from step $4b$, observed at time t.

5. We repeat steps 1 through 5 for a total of 10,000 times to obtain empirical distributions for the R^2 statistics of the two return prediction methodologies.

Table 7 provides the results from this test, ie, the adjusted R^2 and its standard deviation for excess returns of initial maturities ranging for 2 to 10 years, obtained using the Cochrane–Piazzesi and the cointegration factors. While the predictive power obtained using the Cochrane–Piazzesi factor is always higher than what obtained with the cointegration factor, the latter captures an important part of the Cochrane–Piazzesi predictability, with the fraction increasing from 72% to 85% as the investment maturity increases.

As noted above, the explanatory power afforded by the cointegration relationships is always somewhat smaller than the predictive power of the Cochrane–Piazzesi factor. The similarity between the two factors is however striking, as shown in Fig. 6, which displays the Cochrane–Piazzesi and the cointegration return-predicting factors obtained with three (left panel) and five (right panel) forward rates. Although not immediately apparent from the figure, the Cochrane–Piazzesi factor exhibits slightly higher persistence compared to the cointegration return prediction factor. By using the Schwarz information criterion, we have tested for the most parsimonious model of the form AR(p), with $p = 1, 2, \dots, 5$. We have confirmed that all the series follow AR(1) processes: for the three-forward rate case, we find half-lives of 12 and 8 months for the Cochrane–Piazzesi factor for the cointegration factor, respectively; when we use five forward rates, these half-lives estimates become 10 and 6 months, respectively. The Cochrane–Piazzesi and the cointegration factors do not just have similar reversion speeds, but they are also strongly correlated, as shown by the correlation coefficient, which is 0.95 for the return prediction factors generated using three forward rates and 0.93 when five forward rates are used. Both the Cochrane–Piazzesi and cointegration factors therefore display very similar cyclical behaviour. Given the minor differences in their half-life estimates, and the very high degree of synchronicity, their links with macroeconomic variables (such as, e.g., recessions or inflationary bouts), if present, are certainly shared by both approaches. Whether such common links exist is an interesting topic for future investigation. What we can already observe, however, is that, contrary to what we obtained in the case of the slope factor, when we regress against the Cochrane–Piazzesi factor the residuals from a linear fit of the excess returns against the cointegration factor, the R^2 is statistically not different from zero, and neither intercept nor slope are significant: this analysis of residuals indicates that, while the Cochrane–Piazzesi factor contains useful predicting information over and above what is embedded in the slope, it does not offer linear predictability over what is afforded by the cointegration factor.

[14] Equation (3) in Cochrane and Piazzesi (2005b).

Table 7
Adjusted R^2 statistics of return predicting regressions.

	CP (Mean)	CP (Std)	CI (Mean)	CI (Std)
rx^2	0.136	0.070	0.098	0.078
rx^3	0.148	0.068	0.111	0.079
rx^4	0.154	0.068	0.120	0.080
rx^5	0.157	0.068	0.125	0.080
rx^6	0.157	0.069	0.127	0.080
rx^7	0.155	0.069	0.128	0.080
rx^8	0.153	0.070	0.127	0.080
rx^9	0.150	0.070	0.126	0.080
rx^{10}	0.146	0.070	0.125	0.080

The table shows the adjusted R^2 and its standard deviation using the Cochrane–Piazzesi and the cointegration factors (columns labelled 'CP' and 'CI', respectively) across the 10,000 resampled (Gürkaynak and Wright, 2012) data described in Section 3. For each resampled data set the Cochrane–Piazzesi return-predicting regression is run alongside the return-predicting regression using the estimated cointegration relationships for excess returns at maturities $\{2, 3, \ldots, 10\}$. The adjusted R^2 statistics are collected each time the regressions are run, and the resulting mean and standard deviations are reported in the table. CP refers to Cochrane–Piazzesi, and CI refers to the cointegration-based return predictor.

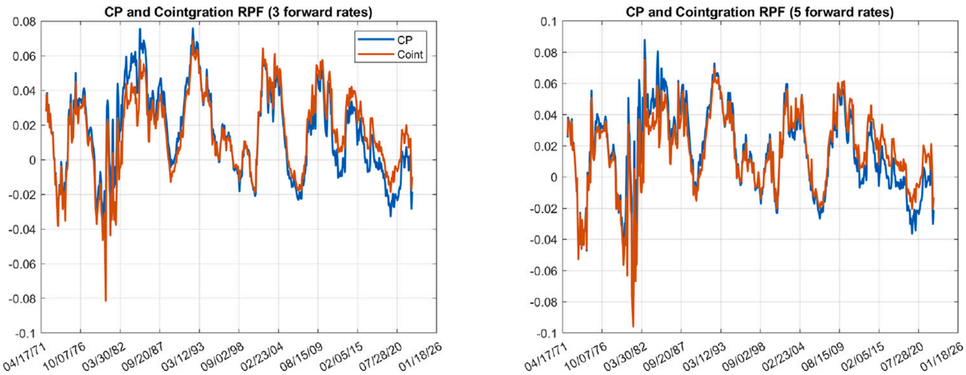

Fig. 6. The left panel shows the return predicting factors using 3 forward rates for the Cochrane–Piazzesi (blue line) and the cointegration model (red line). Same information for 5 forward rates in the right panel.

5.3. Cointegration factors as generators of excess returns

On important piece of our explanation is still missing. To have predictability, it is not enough to find coefficients that turn the $I(1)$ processes for forward rates into a $I(0)$ process. In addition, the resulting stationary process must satisfy at least two further conditions. First, for the resulting stationary process to predict excess returns effectively, it must have a reversion speed similar to the reversion speed of the excess return process. Second, the two processes must be approximately 'in phase'.

It is easy to confirm that the first condition is indeed met. If we assume an $AR(1)$ process of the form $x_{t+1} = \beta + \phi x_t + \epsilon_{t+1}$ for excess returns and for the return-predicting factors, and we measure the auto-regression parameters ϕ for these processes, we find a value of 0.9305 for the average excess return process, of 0.9657 for the slope factor, of 0.9245 for the Cochrane–Piazzesi factor, and of 0.9246 for the cointegration factor. So, the reversion speeds of the process generated by the cointegration factor (and of the slope and Cochrane–Piazzesi factors) are extremely similar to the reversion speed of excess-returns process. What about the second ('phase') condition?

As for the requirement that the mean-reverting processes for the factors should also be 'in phase' with the excess return process, the explanation proceeds in two stages. First, we note that each cointegration relationship describes what we have called a 'mode of attraction', thanks to which forward rates do not stray too far from each other. Second, we argue that these modes of attraction actually *create* the predictable part of excess returns by forcing the forward rates together when they move apart (and, by doing so, we also explain *why* the reversion speeds are so similar).

This can be seen very clearly for the particularly simple case of the slope (that we have interpreted as an approximate cointegration relationship between the first and the last forward rate). In this simple case, the mode of attraction is just the tendency of the yield curve to flatten when very steep (with our definition of slope, this means the tendency of the tenth forward rate to decline when it is 'too far above' the first forward rate), and *vice versa*. This is in essence the excess-return-generating mechanism observed as early as the late 1980s by Fama and Bliss (1987), and shortly after by Campbell and Shiller (1991). In this simple case, it is this mean-revering behaviour of the slope that gives rise to the excess returns: when the tenth forward rate is 'very high', the carry strategy is long the long-dated bond and short the 1-year bond, and therefore benefits from the (predictable) decline in the long-end of the yield curve. We stress that it is because this 'mode of attraction' lends itself to prediction, that also the excess returns are predictable: the predictability of the latter in inherited from the predictability of the former.

As more forward rates are added, visualizing the higher-order modes of attraction becomes more difficult, but the underlying mechanism remains the same: the five-forward-rates Cochrane–Piazzesi and cointegration models simply extend the range of 'modes of attraction' to more complex types of reversion. Indeed, the fact that there is 'extra information' in the more complex modes of attraction explains why, as we discussed in Section 3, the five-forward-rates Cochrane–Piazzesi factor accounts for a significant part of the variability of the residuals from the slope regression (which only 'knows about' one mode of attraction), but not the other way around.

Finally, we note that the interpretation of excess returns being generated by forward rates or yields returning with a characteristic speed of reversion to 'where they should be' is also key to models such as the (Cieslak and Povala, 2015) (see, in particular the interpretation of the Cieslak–Povala 'cycles' presented in Rebonato and Hatano (2022)).[15] Also in the case of the Cieslak and Povala (2015) model, the cyclical reversion of forward rates to 'where they should be' is not only associated with, but actually *generates*, the excess returns. Our findings can therefore be understood as a way of linking, and finding similarities between, apparently very different return-predicting models such as the slope, the Cochrane and Piazzesi (2005a) and the Cieslak and Povala (2015).

6. Conclusions

In our investigation of the Cochrane–Piazzesi model's predictive power for Treasury returns, we have traversed a comprehensive path, examining the model's empirical performance, discussing prevailing explanations for its success, and proposing a novel interpretation grounded in the time-series dynamics of forward rates and their cointegration relationships. Our analysis leads to several key conclusions that not only shed light on the model's effectiveness, but also contribute to the broader understanding of yield curve dynamics and their implications for predicting Treasury returns.

Our empirical analysis unequivocally demonstrates the Cochrane–Piazzesi model's robust out-of-sample predictive capabilities for Treasury returns. By employing a powerful non-parametric resampling technique and using real-world coefficients in simulated environments, we provide compelling evidence that the model's predictive strength is not an artefact of over-fitting, but a genuine feature of its construction.

Contrary to what is often claimed, our research reveals that the specific 'tent' shape of the factor loadings and the four-to-five-year yield spread, previously deemed critical to the model's success, are not indispensable for its predictive prowess. This insight emerges from our analytical exploration and empirical tests, which show that the Cochrane–Piazzesi factor retains its predictive ability even when these elements are absent from the regression models.

At the heart of our contribution is a novel interpretation of why the Cochrane–Piazzesi model excels in predicting Treasury returns. We propose that the model's effectiveness stems from its ability to capture cointegration relationships among forward rates, thereby transforming their non-stationary behaviour into a stationary process that mirrors the characteristics of excess returns. But we go one step further, by suggesting that the 'modes of attraction' associated with the different cointegration relationships actually *generate* the excess returns — and that the predictability of the latter is inherited by the predictability of the former.

Our study has significant implications for both theory and practice. By unveiling the role of cointegration in the model's predictive success, we provide a new lens through which to view the dynamics of the yield curve and its components. Different predicting factors, which at first blush seem to have very little in common, but which are highly correlated (such as the slope, the Cochrane–Piazzesi and cointegration factors), are interpreted as different instantiations of the cointegration relationships we have highlighted.

Our investigation into the Cochrane–Piazzesi model therefore not only reaffirms its status as a powerful tool for predicting Treasury returns, but also advances our understanding of the underlying mechanisms at play.

CRediT authorship contribution statement

Riccardo Rebonato: Software, Investigation, Writing – original draft, Conceptualization, Methodology. **Ken Nyholm:** Writing – review & editing, Formal analysis, Validation, Investigation, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

[15] The expression, applied to forward rates, 'where they should be' has recently been used in the literature – e.g., by Favero et al. (2024) and by Berardi et al. (2021) – with reference to macroeconomic fundamentals like productivity, demographics, and the natural rate of interest. We stress that we do not use the where-they-should-be expression in the same sense, but purely with reference to the cointegration relationships that we identify. Whether the 'where-they-should-be' mechanism in our meaning of the term has some deep links to the same expression in the macroeconomic literature is an interesting question, that we leave open for future investigation.

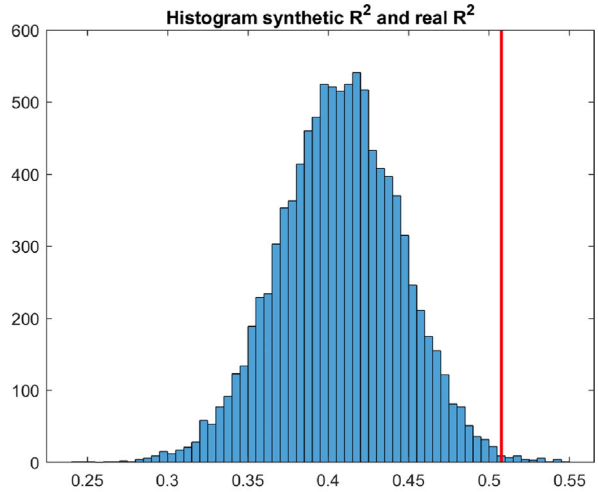

Fig. 7. The real-world coefficient of determination, R^2_{real}, (vertical red line), and the distribution of the resampled R^2 obtained with a block length of 55.

Appendix A. The splicing bias

The resampling procedure by Crump and Gospodinov (2019a) introduces a downward bias in the estimation of the predictive power for excess returns (as measures, for instance, by the R^2 coefficient of determination). The origin of the bias can be easily understood as follows. Consider the case of the estimation of yearly excess returns with monthly data, and of resampling blocks of length $n+1$. Let the original real-world data be time-indexed with an index k, with $k = 1, 2, \dots, N$. Let the ith block contain real-world data time-indexed from time r to time $r+n$, and the $(i+1)$ block contain real-world data time-indexed from time s to time $s+n$.

In the resampled universe, forward rates in block i with time indices $r+n, r+n-1, \dots, r+n-11$ in the real world predict returns for 12-month strategies starting at times $r+n+11, r+n+10, r+n+9 \dots, r+n$, but in the resampled world the forward rates predict excess returns for 12-month strategies starting in block $i+1$ at times $s, s+1, s+2, \dots$, which in the real world were associated with completely different time indices. So, the last eleven predictors of one block are associated with resampled returns built with fewer and fewer yields belonging to that block — the fewer, the more one approaches the end of block r. In particular, the forward rate predictors with time labels $r+n$ predict returns based on rates that are wholly absent in the real world. This creates a downward bias in the resampled predictive power.

We estimate the magnitude of the bias as follows. First we calculate the excess return for the real US data that we have described in Section 3, and the R^2 of the regression of the average excess returns against the Cochrane–Piazzesi return-predicting factor, as described in the text. Next, we resample the same data using blocks of length 50, 55, 60, …, 80. For each block length, we calculate the excess returns, and we regress the resampled excess returns against the resampled Cochrane–Piazzesi factor. Then, for each block length, n, we calculate the percentage difference, $\Delta^{(n)} = \log \frac{R^2_{real}}{R^2_{synth}(n)}$, between the resampled ('synthetic') R^2_{synth} and the real-world, R^2_{real}, R^2 coefficients. Finally we regress the percentage difference against the block length, n:

$$\Delta^{(n)} = \alpha + \beta n + \epsilon \tag{8}$$

We expect the resampled R^2_{synth} to be lower than the real-world R^2_{real}, and the difference to increase as the block length decreases (ie, as the fraction of 'inconsistent' returns increases). Fig. 7 shows the real-world coefficient of determination, R^2_{real}, (vertical red line), and the distribution of the resampled R^2 obtained with a block length of 55. When we run regression (8), we obtain $\alpha = 0.3138$, $\beta = -0.0021$, with $R^2 = 0.8788$, a p value of 0.0018, $tstats = 14.51(\alpha), -6.0125(\beta)$, showing that the block length has a clear influence on the degree of predictability, and that it introduces a downward bias (negative β coefficient).

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